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```
% ECE 2409-001
```

Nicholas Florio

Exam 2

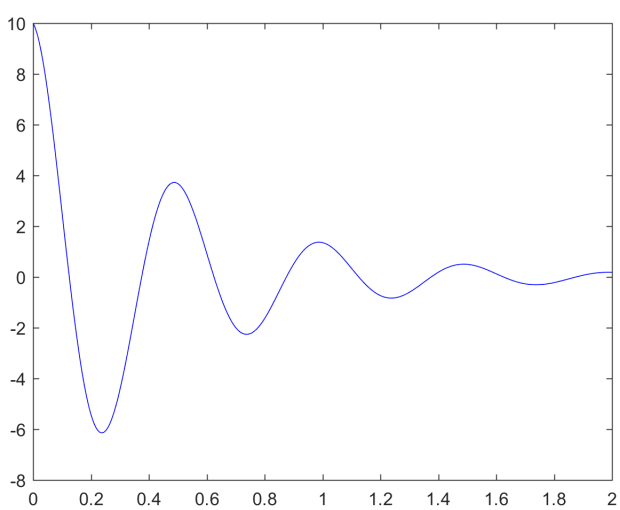
```
clc;close all;clear
datestr(now, 21)
```

```
ans =

    'Nov.20,2017 16:51:33'
```

Problem 1

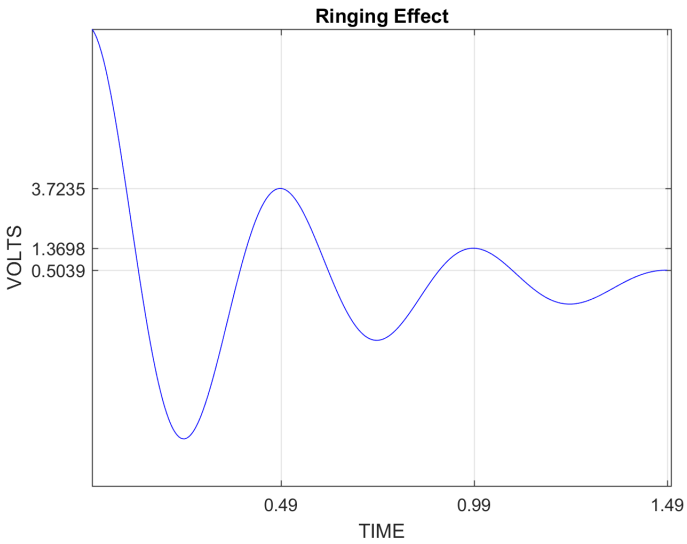
Part A

```
t= [0:1/300:2];
c=10*exp(-2*t).*cos(4*pi*t);
plot(t,c,'b');
```



Part B, D, E

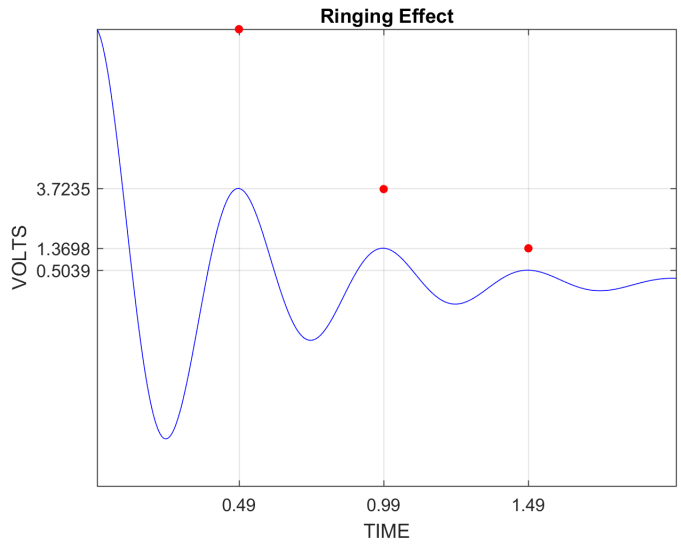
```
t= [0:1/300:1.49];
c=10*exp(-2*t).*cos(4*pi*t);
plot(t,c,'b');
xlabel('TIME');
ylabel('VOLTS');
title('Ringing Effect');
grid
set(gca, 'ytick', [.5039,1.3698,3.7235]);
set(gca, 'xtick', [0.49:.5:1.49]);
```



Part C

```
t= [0:1/300:2];
c=10*exp(-2*t).*cos(4*pi*t);
plot(t,c,'b');
xlabel('TIME');
ylabel('VOLTS');
title('Ringing Effect');
grid
set(gca, 'ytick', [.5039,1.3698,3.7235]);
set(gca, 'xtick', [0.49:.5:1.49]);
remax= [c(1:length(c)*1.49/2/3);c(length(c)*1.49/2/3+1:length(c)*1.49/2/3*2);c(length(c)*1.49/2/3*2+1:length(c)*1.49/2)] ;
remax = max(remax')
xmax= [0.49,.99,1.49]
hold on
plot(xmax,remax,'r.', 'markersize', 16);
```

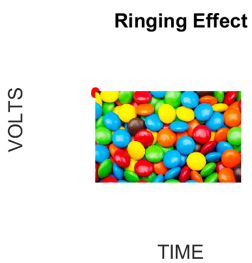
```
Warning: Integer operands are required for colon operator when
used as index
Warning: Integer operands are required for colon operator when
used as index
Warning: Integer operands are required for colon operator when
used as index

remax =

   10.0000    3.7002    1.3667


xmax =

    0.4900    0.9900    1.4900
```



Problem 2

Part A

```
meltsinyourmouth= imread('mandm.jpg');
imshow(meltsinyourmouth)
```



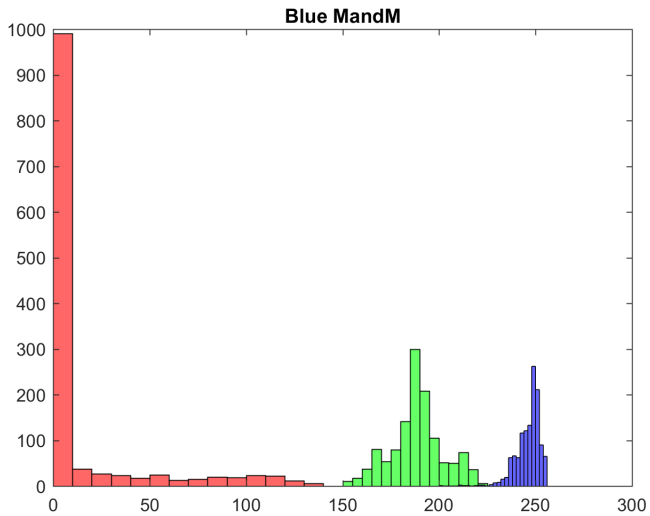
Part B

```
openfig('mandmfigureplot.fig');
```

Part C

```
BLUE= meltsinyourmouth(80:129,48:72,:);
R=BLUE(:,:,1);
G=BLUE(:,:,2);
B=BLUE(:,:,3);
figure
histogram(R, 'facecolor', 'r')
hold
histogram(G, 'facecolor', 'g')
histogram(B, 'facecolor', 'b')
title('Blue MandM')
```

```
Current plot held
```



Part D and E

```
R2=meltsinyourmouth(:,:,1);
G2=meltsinyourmouth(:,:,2);
B2=meltsinyourmouth(:,:,3);
BLUE=[G2<=215 & B2>=215 & R2<=215];

figure
subplot 121;
imshow(BLUE);
subplot 122;
imshow(meltsinyourmouth);
```

Part F

```
 pictureBlue=imread('mandm.jpg');
R=pictureBlue(:,:,1);
G=pictureBlue(:,:,2);
B=pictureBlue(:,:,3);

Blue=(R<=215 & G<=215 & B>215);
Place=find(Blue==0);
R(Place)=0;
G(Place)=0;
B(Place)=0;

pictureBlue(:,:,1)=R;
pictureBlue(:,:,2)=G;
pictureBlue(:,:,3)=B;
figure
imshow(pictureBlue)
title('Blue Colored MandMs')
 [centersblue,radiiblue]=imfindcircles(pictureBlue, [30 80], 'Sensitivity',0.98);
Blueamount=length(centersblue)
```

```
figure
imshow(pictureBlue)
title('Blue MandMs Counted')
centerBlue=centersblue(1:Blueamount,:);
radiiBlue=radiiblue(1:Blueamount,:);
viscircles(centerBlue, radiiBlue, 'edgecolor', 'b');
```

```
Blueamount =

    16
```

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